$h : Loc \longrightarrow Obj$ $\qquad$ $sto : Var \longrightarrow [Loc]$

$Obj : Prop \longrightarrow Loc$

$analyse(If(e)\{s_1\}else\{s_2\}, h, \rho) =$

$\quad let\ \rho' = copy\ \rho$

$\quad let\ \rho_1 = analyse(s_1, h, \rho)$

$\quad let\ \rho_2 = analyse(s_2, h, \rho')$

$\quad return\ merge(\rho_1, \rho_2)$

```
merge ( f1 , f2 ) {
    let f = new f
    for each var x in dom(f1) ∩ dom(f2) .
        f[x] = f1[x] ∪ f2[x]
    for each var x in dom(f1) - dom(f2)
        f[x] = f1[x]
    for each var x in dom(f2) - dom(f1)
        f[x] = f2[x]
```

0

1

if (o.x > 0) {

2 z = o.v

3 o.x = z;

} else {

o.x = 3 4

}

eval (o.x)

o.x = 3

eval (o.x)

0 $h_0 = \emptyset$, $\rho_0 = \emptyset$

1 $h_1 = [l_0 \mapsto \{x : l_1\}, l_1 \mapsto \{\}]$, $\rho_1 = [o \mapsto [l_0]]$

Then

2 $h_2 = [l_0 \mapsto \{x : l_1, w : l_2\}, l_1 \mapsto \{\}, l_2 \mapsto \{\}]$

$\rho_2 = [o \mapsto [l_0], z \mapsto [l_2]]$

3 $h_3 = [l_0 \mapsto \{x : l_1, w : l_2\}, l_1 \mapsto \{\}, l_2 \mapsto \{\},$

$l_3 \mapsto \{x : l_2, w : l_2\}]$,

$\rho_3 = [o \mapsto [l_3], z \mapsto [l_2]]$

4 $h_4 = [l_0 \mapsto \{x : l_1, w : l_2\}, l_1 \mapsto \{\}, l_2 \mapsto \{\},$

$l_3 \mapsto \{x : l_2, w : l_2\}, l_4 : \{x : l_5, w : l_2\}, l_5 \mapsto \{\}]$

$\rho_4 = [o \mapsto [l_4]]$

0

1

if (o.x > 0) {

2 z = o.v

3 o.x = z;

} else {

o.x = 3 4

}

eval (o.x) 5

o.x = 3

eval (o.x)

3 $h_3 = [l_0 \mapsto \{x : l_1, w : l_2\}, l_1 \mapsto \{\}, l_2 \mapsto \{\},$

$l_3 \mapsto \{x : l_2, w : l_2\}],$

$\rho_3 = [o \mapsto [l_3], z \mapsto [l_2]]$

4 $h_4 = [l_0 \mapsto \{x : l_1, w : l_2\}, l_1 \mapsto \{\}, l_2 \mapsto \{\},$

$l_3 \mapsto \{x : l_2, w : l_2\}, l_4 : \{x : l_5, w : l_2\}, l_5 \mapsto \{\}]$

$\rho_4 = [o \mapsto [l_4]]$

5 $h_5 = h_4$

$\rho_5 = [o \mapsto [l_3, l_4], z \mapsto [l_2]]$

$\downarrow$